MILITARY SPECIFICATION

MICROCIRCUIT, DIGITAL, CMOS (LOGIC)

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements and provides quality assurance provisions for fabrication and packing of a complementary, metal oxide, semiconductor (CMOS) integrated circuit used in the Mine, antitank: BLU-91/B, and Mine, antipersonnel BLU-92/B.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-38510 - Microcircuits, General Specification for
MIL-M-55565 - Microcircuits, Preparation for Delivery of

STANDARDS

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microcircuits
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

9332000 - Integrated Circuit, Digital, CMOS (Logic)

FSC: 1345

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, drawings -d publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.3 Order of precedence In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Device classification. The microcircuit (device) shall conform to the product assurance level class B of MIL-M-38510 and as specified herein.

3.2 Device. The device shall comply with all requirements specified on Dwg. 9332000 and with all requirements specified in applicable specifications and standards.

3.3 Design and construction The design and construction shall comply with the requirements of MIL-M-38510 for class B devices and 3.2 herein.

3.3.1 Glassivation. The microcircuit shall be glassivated with silicon dioxide to render it impervious to a contaminated environment. The glassivacion shall cover all electrical conductors except the bonding pads.

3.4 Environmental and mechanical Devices to be delivered under this specification shall be capable of meeting the following MIL-STD-883 environmental and mechanical tests when the devices are subjected to these tests in accordance with this specification.

Environmental tests	Method	Conditions
Moisture resistance	1004	
Steady state life	1005	D, (1000 hrs @ 125°C)
Temperature cycling	1010	C, 20 cycles
Seal (fine and gross)	1014	(A & C, respectively)
Stabilization bake	1008	C
Burn in	1015	D, 160 hrs @ 125°C
Mechanical shock	2002	G
Solderability	2003	
Lead integrity	2004	B2
Resistance to solvents	2015	
Vibration, variable frequency	2007	A

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Environmental tests	Method	Conditions
External Visual	2009	
Internal Visual	2010	B
Constant acceleration	2001	E, Y_1, orientation only
Bond strength	2011	D
Physical dimensions	2016	

3.5 Electrical requirements. The device shall have the following electrical characteristics.

3.5.1 Operating ratings: See Table I.

TABLE I. Operating Ratings.

CHARACTERISTIC	SYMBOL	RATING	UNIT
DC Supply Voltage	V_{DD}	-0.5 to 7.5	Volts
All Inputs	V_{IN}	V_{IN} ≤ V_{DD}	Volts
Temperature			
Operating 1/	$T_{A(OP)}$	-55 to +80	°C
Storage	$T_{A(STG)}$	-55 to +125	°C

1/ Device shall be capable of burn-in at +125°C. (See 4.5.1.7)

3.5.2 Performance characteristic. The device shall meet the performance characteristics of Table II.

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TABLE II. ELECTRICAL PARAMETERS.

CHARACTERISTICS	PINS	V_{DD}	T_A 1/	MIN	MAX	UNIT	2/	3/
Leakage Current	16	7.5 v	25		3.0	uA	3.5.2.1	4.5.4
Low		7.5 v	-55/80		15.0	uA		
Leakage Current	16	7.5 v	25		20.0	uA	3.5.2.2	4.5.5
High		7.5 v	-55/80		30.0	uA		
Input Current	1,4,7,9,12, 13,14,15	7.5 v	25		100.0	nA	3.5.2.3	4.5.6
Input Current	1		-55/25	10.0		uA	3.5.2.4	4.5.7
	14, 15		-55/25	0.3	6.0	uA		
	13		-55/25	0.3	6.0	uA	3.5.2.5	4.5.8
Output Low	2	5.0 v	25/80	5.0		uA	3.5.2.6	4.5.9
	5, 6	5.0 v	25/80	200.0		uA	3.5.2.7	4.5.10
	10, 11	5.0 v	25/80	10.0		uA	3.5.2.8	4.5.11
Output High	3	5.0 v	25/80	5.0		uA	3.5.2.9	4.5.12
	5, 6	5.0 v	25/80	1.5		mA	3.5.2.10	4.5.13
	10, 11	5.0 v	25/80	10.0		uA	3.5.2.11	4.5.14
Switching point (OSC)	2, 3, 9		-55/80	2.5	5.0	V	3.5.2.12	4.5.15
Switching point (LVD)	4	5.0 v	25	1.1	2.5	V	3.5.2.13	4.5.16
LVD Temperature Coefficient							See 3.5.5	4.5.17
Slow Rise	7		-55/80	1.0		mSec	3.5.2.14	4.5.19
Oscillator Functional			25		60.0	uA	3.5.2.15	4.5.20
Current			-55/80		90.0	uA		
Slow Power Up			-55/80	4.0	500.0	mSec	3.5.2.16	4.5.21
Sequential Test							See 3.5.3	4.5.22

1/ All temperatures specified in °C.

2/ Requirement paragraph (assigned in this table).

3/ Test Method paragraph.

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3.5.3 Sequential operations. When the component is connected as shown in Figure 3 and the waveforms shown in Figures 4 thru 10 are applied to terminals 1, 2, 3, 4, 7, 9, 12, 13, 14, 15, operation shall be as described by the output waveforms at Figures 4 thru 10 when tested as specified in 4. 5.22

3.5.4 Oscillator function. When the component is connected as shown in Figure 1, it shall function as an oscillator with the following additional requirements:

(a) Pins 9, 10 and 11, When the temperature varies over the operating range (-55°C to 80°C) and the voltage varies from 3.0 to 4.5 volts, the output period shall not shift more than ±5% from the period at +25°C and V_B = 4.0 volts.

(b) Pins 2 and 3. When the temperature varies over the operating range (-55°C to +80°C) and the voltage varies from 5 to 7.5 volts, the output period shall not shift more than + 5% from the period at 25°C and V_B = 6.4 volts.

These requirements shall be tested for as specified in 4.5.18.

3.5.5 LVD temperature coefficient. The LVD temperature coefficient of pin 4, switching point shall not exceed 2.58 millivolts per C when measured over the temperature range of -40°C to 63°C when tested as specified in 4.5.17.

3.5.6 Screening. All devices to be delivered in accordance with this specification shall have been subjected to the screening procedures as specified in MIL-STD-883 Method 5004 for a class B device, and as specified herein (see 4.4.2.1).

3.5.6.1 Burn-in. The devices shall be subjected to the burn-in test as specified in MIL-STD-883 Method 1015 and 4. 5.23 herein.

3.5.7 Steady state life. The devices shall be subjected to the steady state life test of MIL-STD-883 Method 1005 condition D and 4.5.24 herein.

3.6 Device marking, The device shall be marked with the information specified in 3.6.1 thru 3.6.6, and the primary markings (index point, part number, and inspection lot identification code) shall be located on the top surface of the package. The ocher markings may be placed in any suitable location so as to perform their functions and not interfere with the primary markings.

3.6.1 Index point. See 3.2.

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3. 6.2 Part number. The complete part number shall be as follows:

9332000X

where 9332000 is the microcircuit Drawing number and X the drawing revision letter, if applicable.

3.6.3 Inspection lot identification code. The inspection lot identification code marking shall be in accordance with MIL-M-38510.

3.6.4 Metallization identification. The device shall be marked with a date code that can be used (with quality assurance program records) to trace individual device packages (sealed) and to a specific metallization run (batch).

3.6,5 Manufacturer's designating symbol . The manufacturer's designating symbol marking shall be in accordance with MIL-M-38510.

3.6.6 Other markings. The vendor may add additional markings, for his own use, on the bottom surface of the package provided that these markings do not interfere with the required markings,

3.7 Data requirement. Data shall be recorded and retained for all tests required by this specification in accordance with 4.5.22.

3.8 Process controls. All process controls essential for the manufacturing of microcircuits to assure a reliable end item shall be submitted to the designated agency specified in 6.5 herein, prior to the start of fabrication.

3.9 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.10 Country of origin. The device shall be manufactured and sold by a qualified producer in the United States or in another country which is a party co one of the existing international agreements ratified by the United States, such as the NATO Electronic Parts, and the "United States/Canada/Australia Agreement for Qualification of Products for Non-resident Manufacturers.

3.11 Workmanship. (See MD-M-38510).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-M-38510 shall apply-and are hereby made part of this detail specification. The supplier is responsible for the performance of all inspection requirements as specified herein, and the supplier may use his own

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or any other facilities suitable for the performance of the tests and inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to witness or to perform any of these tests and inspections set forth in this specification where such test and inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. The Government reserves the right to audit the data resulting from the manufacturer's performance of these tests and inspections.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

c. Screening

4.3 First article Inspection,

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article inspection lot size after screening shall consist of fifty (50) devices, minimum.

4.3.2 Inspection to be performed. First article inspection shall be in accordance with the applicable requirements of MIL-M-38510 and as modified by this specification, except that the procuring agency shall be the Qualifying activity. First article requirements shall consist of inspections specified in subgroups "a" through "f" listed in TABLE III. First article items my be subjected by the qualifying activity (Government) to all of the examinations and tests specified in this specification and to any requirements of the applicable drawings.

a. Screening . Screening shall be performed on sll devices in accordance with the inspection flow specified in 4.4.2.1.

4.3.3 Rejection If any device fails to comply with any of the applicable requirements the first article sample shall be rejected. The Goverment reserves the right to terminate its inspection upon any failure of a device in the sample to comply with any of the stated requirements.

4.3.4 Requalification. The device shall be subjected to another First Article as stipulated in the contract if:

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The contractor's manufacturing te quniques or process controls which affect the form, fit, function or internal geometry of the device are revised by either the contractor or one of his sub-contractors supplying parts or assemblies.

b. The contractor relocated his assembly facilities.

c. The contractor uses a new sub-contractor for material, parts, or assembly.

d. A change occurs in devices's material, performance or assembly.

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TABLE III. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63530(AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
	MICROCIRCUIT, DIGITAL, CMOS				9332000
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Screening	1/		3.5.6	4.4.2.1
	Subgroup a	25			
	Electrical parameters @ -50° ± 5°C			3.5.2	Table II, 4.4.3.1
	Electrical parameters @ +75° ± 5°C			3.5.2	Table II, 4.4.3.1
	Oscillator function			3.5.4	4.5.15
	LVD temperature coefficient			3.5.5	4.5.12
	Subgroup b	25			
	Mechanical shock			3.4	MIL-STD-883, Method 2002, G
	Vibration, variable frequency			3.4	MIL-STD-883, Method 2007, A
	Moisture resistance			3.4	MIL-STD-883, Method 1004
	Seal (fine leak)			3.4	MIL-STD-883, Method 1014, A
	Seal (gross leak)			3.4	MIL-STD-883, Method 1014, C
	External visual			3.3,3.4, 3.6,3.10	MIL-STD-883, Method 2009
	Electrical parameters @ 25°C ± 5°C			3.5.2	Table II, 4.4.3.1
	Country of origin			3.10	Visual
	Subgroup c	25			
	Resistance to solvents	3		3.4	MIL-STD-883, Method 2015
	Bond strength (bond pulls 2/)			3.4	MIL-STD-883, Method 2011, D

NOTES

See page 2. FIRST ARTICLE Sample of 50 Devices: 25 for groups a-f; 25 for group g.

DESC-QA Form 150 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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TABLE III. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63530(AR)

PARAGRAPH	TITLE: MICROCIRCUIT, DIGITAL, CMOS		SHEET 2 OF 3		DRAWING NUMBER: 9332000
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Subgroup d	25			
	Solderability (leads inspected 3/)			3.4	MIL-STD-883, Method 2003
	Subgroup e	2			
	Lead integrity			3.4	MIL-STD-883, Method 2004, B2
	Seal (fine leak)			3.4	MIL-STD-883, Method 1014, A
	Seal (gross leak)			3.4	MIL-STD-883, Method 1014, C
	Subgroup f	25			
	Steady state life			3.5.7	4.5.18
	Electrical parameters @ 25°C ± 5°C			3.5.2	Table II, 4.4.3.1
	Notes: 1/ Devices which fail any test criteria in the screening sequence shall be removed from the First Article Inspection lot. The contractor should choose a First Article Inspection lot size that is adequate for the needs of the First Article Inspection. 2/ The sample size is the number of bond pulls selected from a minimum number of 10 devices. 3/ The sample size is the number of leads inspected: In no case shall less than 3 devices be used to provide the number of required leads				

NOTES:

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TABLE III. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63530(AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
	MICROCIRCUIT, DIGITAL, CMOS		3 OF 3	9332000
				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH / PARAGRAPH REFERENCE /INSPECTION METHOD
	4/ All electrical parameters shall be read and recorded as go/no go data. 5/ The same sample units shall be subjected to all the tests in a subgroup. 6/ Devices used in subgroup a shall be used in subgroup b. Devices used in subgroups a and b may be used in subgroups c, d and e. 7/ Tests within each subgroup shall be performed in the sequence shown.			

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lot formation shall be as defined in MIL-M-38510 for Class B devices.

4.4.2 Examinations.

a. Quality conformance inspection. Quality conformance inspection shall consist of the inspections specified for Group A and Group B and when required Group C. Quality conformance inspection requirements shall be in accordance with the Classification of Defects and Test paragraph 4.4.2.1 through 4.4.2.4.

b. Inspection requirements. Inspection requirements for this device shall be in accordance with the requirements of MIL-M-38510 and MIL-STD-883, Method 5005 for Groups A, B, and C tests except as modified herein. See 4.4.2.1 thru 4.4.2.4.

c. Screening shall be performed on all devices in accordance with the inspection flow specified in 4.4.2.1.

d. Sampling. Inspection samples of devices required by this specification shall be randomly selected from each inspection lot in accordance with Appendix B-1 of MIL-M-38510.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63530(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Microcircuit, Digital, CMOS			9332000
				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST SCREENING: (3.4, 3.5.6)	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Internal visual (Pre-cap)		100%	3.5.6	MIL-STD-883, Method 2010, B
	High temperature storage (post seal stabilization bake)		100%	3.4,3.5.1 3.5.6	MIL-STD-883, Method 1008, C
	Temperature cycling		100%	3.5.6	MIL-STD-883, Method 1010, C
	Constant acceleration		100%	3.5.6	MIL-STD-883, Method 2001, E Y1 orientation only
	Seal (fine leak)		100%	3.5.6	MIL-STD-883, Method 1014, A
	Seal (gross leak)		100%	3.5.6	MIL-STD-883, Method 1014, C
	Electrical parameters @ 25°C (pre burn-in)		100%	3.5.6	Table II, 4.4.3.1 herein
	Burn-in		100%	3.5.6.1	4.5.17 herein
	Electrical parameters @ 25°C (post burn-in)		100%	3.5.2	Table II, 4.4.3.1 herein
	External visual		100%	3.5.6, 3.3,3.4, 3.5.6,3.6 3.10	MIL-STD-883, Method 2009
	Country of origin		100%	3.11	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63530(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Microcircuits, Digital, CMOS		9332000
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST GROUP A: (see 4.4.3.2, 4.4.3.3)	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Electrical parameters @ -55°C ± 5°C		LTPD 5%	3.5.2	Table II, 4.4.3.1 herein
	Electrical parameters @ +80°C ± 5°C		LTPD 5%	3.5.2	Table II, 4.4.3.1 herein
	Oscillator function		LTPD 15%	3.5.4	4.5.15
	LVD temperature coefficient		LTPD 5%	3.5.5	4.5.12

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Microcircuit, Digital, CMOS			9332000
				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
	GROUP B: (see 4.4.3.4)				
	Subgroup 1				
	Physical dimensions	4	0-1	3.2,3.4	MIL-STD-883, Method 2016
	Subgroup 2				
	Resistance to solvents	3	0-1	3.4	MIL-STD-883, Method 2015
	Bond strength (bond pulls 1/)		LTPD 15%	3.4	MIL-STD-883, Method 2011, D
	Subgroup 3				
	Solderability (leads inspected 2/)		LTPD 15%	3.4	MIL-STD-883, Method 2003

NOTES: 1/ The LTPD sample size is the number of bond pulls selected from a minimum number of 10 devices. 2/ The LTPD sample size is the number of leads inspected: In no case shall less than 3 devices be used to provide the number of leads required.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Microcircuit, Digital, CMOS	1 OF 1	9332000
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST GROUP C: (see 4.4.3.5)	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Subgroup 1				
	Steady state life		LTPD 5%	3.5.7	4.5.18
	Electrical parameters @ 25°C ± 5°C		LTPD 5%	3.5.2	Table II, 4.4.3.1 herein
	Subgroup 2				
	Mechanical shock		LTPD 15%	3.4	MIL-STD-883, Method 2002, G
	Vibration, variable frequency		LTPD 15%	3.4	MIL-STD-883, Method 2007, A
	Seal (fine leak)		LTPD 15%	3.4	MIL-STD-883, Method 1014, A or B
	Seal (gross leak)		LTPD 15%	3.4	MIL-STD-883, Method 1014, C
	External Visual		LTPD 15%	3.3,3.4 3.6,3.10	MIL-STD-883, Method 2009
	Electrical parameters @ 25°C ± 5°C		LTPD 15%	3.5.2	Table II, 4.4.3.1 herein
	Packaging		2/	5.1	MIL-M-55565
	Packing		2/	5.1	MIL-M-55565
	Marking		2/	5.2	MIL-M-55565 & MIL-STD-1188

NOTES 1/ The same sample units shall be subjected to all the tests in a subgroup.
2/ Sampling and inspection as per MIL-M-55565.

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4.4.3 Testing. Testing is described in the first article and quality conformance inspection "Classification of Defects and Tests" forms and below.

4.4.3.1 Electrical parameters @ 25° ± 5°C testing. Each device shall be tested ior compliance with the requirements listed in Table II. Any device classified as defective shall be removed from the lot.

4.4.3.2 Testing sequence. The order of group testing shall be A, B, and C: Only lots that passed Group A tests shall be submitted for Group B testing, and only lots that passed Group B tests shall be submitted for Group C testing.

4.4.3.3 Group A testing. Random samples from each inspection lot shall be selected, in accordance with this specification, and tested at the minimum and maximum temperatures for compliance with the requirements listed in Table II. An inspection lot which fails the Group A test shall be classified as defective until all the defective devices within the lot are identified and removed by subjecting all the. devices in the lot to the Group A test.

4.4.3.4 Group B testing, Random samples from each inspection lot that passed Group A tests shall be selected and tested for compliance with the requirements of the Group B tests in accordance with this specification. An inspection lot shall not be shipped if the devices of any subgroup of the Group B test fail to comply with all of the specified requirements.

4.4.3.5 Group C testing. Group C tests shall be performed on a continuing schedule of at least once every ninety (90) days while the device is being produced. Randam samples from one inspection lot that passed Group B tests shall be selected and tested for compliance with the requirements of the Group C tests in accordance with this specifications. Further inspection Lots shall not be shipped if the devices of any subgroup of the Group C tests fail to comply with all of the specified requirements.

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is identified in the Classification of Defects and Tests Forms and in 4.5 herein. The contractor shall submit for approval inspection equipment designs, including alternate equivalent design whit'n he may desire to use, in accordance with the terms of the contract. See MIL-M-38510 and 6.3 herein.

4.5 Test methods, circuits and procedures.

4.5.1 Alternate methods, circuits and procedures. Unless otherwise stated in the specific test method, the methods and circuits shown are given as basic measurement method. For alternate methods, circuits and procedure the manufacturer shall demonstrate to the procuring activity that alternates which he may desire and propose to use are equivalent and give results within the desired accuracy or measurement (see 6.3).

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4. 5.2 Defective. Any device failing to meet the applicable requirements in the following tests shall be classed and labeled defective and segregated from the inspection lot.

4.5.3 Test conditions. Unless otherwise specified all examinations and tests shall be made under the following conditions.

Temperature	25°C
Humidity	10 to 70 percent RH
Pressure	29 + 3 inches Hg
Voltage & Currents	± 2%
Temperature (ALL) Tolerance	± 5°C

4.5.4 Leakage current low. Measure the leakage current into pin 16 after a power up reset with the inputs connected as shown in Figure 2. by device falling to meet the requirements of 3.5.2.1 shall be classed defective.

4.5.5 Leakage current high. Pause at bit time 2768 of functional test sequence #7 as shown in Figure 10 and measure the current into pin 16. The logic state is defined by the functional test for this measurement Any device failing to meet the requirements of 3.5.2,2 shall be classed defective.

4.5.6 Input current. Connect pins 4, 7, 9, 12 and 13 to V_{DD} and measure the current into each pin, connect pins 1, 4, 7, 9, 12, 14 and 15 to V_{SS} (ground) and measure the current out of each pin. Any device failing to meet the max current requirement of 3.5.2.3 shall be classed defective. (Non-destructive test).

4.5.7 Input current. Measure the current into pin 1, 14 and 15, in turn with the pin under test held at V_{DD}. Any device failing to meet requirements of 3.5.2.4 shall be classed defective. (Non destructive test).

4.5.8 Input current. Measure the current out of pin 13 with pin 13 held at V_{SS} (ground). Any device failing to meet the requirements of 3.5.2.5 shall be classed defective. (Non-destructive test).

4.5.9 output low. Force each of pins 2 and 3 to V_{DD} through 5K ohms, then measure the current into pin 2 with pin 2 held at a voltage of 0.1 volts. Any device failing to meet the requirements of 3.5.2.6 shall be classed defective. (Non-destructive test).

4.5.10 Output low. After a power up reset measure the current into each pin with each pin held at 1.0 volts. Any device failing to meet the requirements of 3.5.2.7 shall be classed defective. (Non-destructive test).

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4.5.11 Output low. Measure the current into pin 10 with pin 9 connected to V_{SS} and pin 10 held at 0.1 volts. Measure the current into pin 11 with pin 9 connected to V_{DD} and pin 11 held at 0.1 volts. Any device failing to meet the requirements of 3.5.2.8 shall be classed defective. (Non-destructive test).

4.5.12 Output high. Force each pin 2 and 3 to V_{SS} through 5K ohms, then measure the current out of pin 3 with pin 3 held at a voltage of V_{DD} - 0.1 volts. by device failing to meet the requirements of 3.5.2.9 shall be classed defective. (Non-destructive test).

4.5.13 Output high. Select appropriate points in the functional test to pause and measure the output current drive of pin 5 and then pin 6. Measure the current out of each pin each held at 2.0 volts. Any device failing to meet the requirements of 3.5.2.10 shall be classed defective. (Non-destructive test).

4.5.14 Output high. Measure the current out of pin 10 with pin 9 connected to V_{DD} and pin 10 held at V_{DD} - 0.1 volts. Measure the current out of pin 11 with pin 9 connected to V_{SS} and pin 11 held at V_{DD} - 0.1 volts. Any device failing to meet the requirements of 3.5.2.11 shall be classed defective. (Non-destructive test).

4.5.15 Switching point. Set pin 9 at 2.5 volts and check to see that pin 10 output is less than 1.5 volts. Set pin 9 at the specified maximum voltage and check to see that pin 10 output voltage is greater than V_{DD} - 1.5 volts. Drive pins 2 and 3 through SK ohm resistors to V_{SS} and then set the pin 3 drive input at the specified maximum voltage. The pin 3 P channel transistor should be “on” and the actual pin 3 voltage will differ from the drive input by at least 1.0 volts. Set the pin 2 drive input at the specified minimum voltage and verify that the pin 3 P channel is still “on” (that is no change at pin 3). Set the pin 2 drive input at the specified maximum voltage and verify that the pin 3 P channel is “off”. The N channel transistor on pin 2 should be “on”. Drop the pin 2 drive input to the specified minimum voltage and the actual pin 2 voltage should differ from the drive input by at least 1.0 volts. Set-the pin 3 drive input at the specified minimum voltage and verify that the pin 2 N channel is now "off". Any device failing to meet the requirements of 3.5.2.12 shall be classed defective. (Non-destructive test).

4.5.16 Switching point. This test is run in conjunction with functional test sequence #3 Pause at the point where pin 4 is to be switched from a “l” to a “0”. Set pin 4 at the specified maximum voltage and verify that pin 6 has not switched. Reduce pin 4 to the specified minimum and determine the voltage that pin 6 switches. Any device failing to meet the requirements of 3.5.2.13 shall be classed derective. (Non-destructive test).

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4.5.17 LVD temperature coefficient. While at the temperature extremes specified in 3.5.5 determine the switching point of pin 6 by varying" the voltage on pin 4 from maximum to minimum. Determine the change in switching point voltage per 'C from the value obtained in 4.5,16. Any device failing to meet the requirements of 3.5.5 shall be classed defective. (Non-destructive test).

4.5.18 Oscillator function test. The device shall be connected as shown in Figure 1 and tested for the requirements as specified in 3.5.4. Any device failing to meet the requirements of 3.5.4 shall be classed defective. (Nou-destructive test).

4.5.19 Slow rise. The pin 7 slow rise performance can be verified by rerunning functional test sequence #1 and switching pin 7 through a RC delay whose time constant is greater than or equal to the specified minimum time. Any device failing to meet the requirements of 3.5.2.14 shall be classed defective. (Non-destructive test).

4.5.20 Oscillator functional current. With the circuit connected as shown in Figure 1 measure the average current used with both oscillators running . Any device failing to meet the requirements of 3.5.2. 15 shall be classed defective. (Non-destructive test).

4.5.21 slow power up. Verify the slow power up reset maximum limit by powering the device through a RC delay with a time constant greater than or equal to the maximum time. Verify the slow power up reset minimum time with any power up which occurs in less than the specified minimum time. This test may be performed during any of the seven (7) functional test sequences. Any device failing to meet the requirements of 3.5.2.16 shall be classed defective. (Non-destructive test).

4.5.22 Sequential test. The sequential test consist of seven (7) functional test sequences defined in Figures 4 through 10. Tests 1-6 are to be run with V_B = 5.0 volts, and test 7 with V_B = 7.5 volts. The circuit must be testable with a bit time (time between state changes) of 10 microseconds. All input low logic levels (VIL) shall be 1.5 volts. All input high levels (VIII) shall be V_{DD} - 1.5 volts. All output low levels (VOL) shall be 1.0 volts or less and output 'nigh levels (VOH) shall be V_{DD} 1.5 volts or greater. Any device failing to meet the requirements of 3.5.3 shall be classed defective (Non-destructive test).

4.5.23 Dynamic burn-in test. All devices shall be subjected to the dynamic burn-in screen in accordance with MIL-STD-883, Method 1015, Condition D (160 hours minimum at 125°C + 5°C) and specified herein. For this test the device shall be connected as shown in Figure 11. Any device failing to meet the requirements of 3.5.6.1 shall be classed defective. (Non-destructive test).

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4.5.24 Steady state life. Devices shall be subjected to the steady state life test in accordance with MIL-STD-883, Method 1005, Condition D and as specified herein, Test duration shall be 1000 hours minimum at 125°C ± 5°C. For this test the device shall be connected as shown in Figure 11, the same test set up as for the burn-in test. Electrical parameters shall be measured before, during and after the steady state life test in accordance with Method 1005 and Table II. my device failing to meet the electrical parameters requirements of Table II shall be classed defective. (Non-destructive test).

4.5.25 Data requirements. The contractor shall submit a quality inspection test, demonstration and evaluation report giving all detailed inspection and test results in accordance with Data Item Description DI-R-1724, test report format shall be submitted to the technical agency listed in 6.5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Devices to be delivered under this specification shall be prepared for delivery in accordance with the requirements of MIL-M-38510, Level C. Electrostatic discharge (ES) protection in accordance with the requirements of MIL-M-55565 shall be provided.

5.2 Marking. Marking shall be in accordance with MIL-STD-1188.

5.3. Shipping. When microcircuit circuit packages of more than one lot are shipped together, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The device covered by this specification is intended for use in U.S. Army Ordnance.

6.2 Ordering data. The procurement documents shall specify the following:

(a) Title, number and date of this specification.

(b) Provisions for submission of first article samples.

(c) Quantity required and delivery schedules.

(d) Requirement for certification of conformance for each lot or shipment of product.

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6.3 Submission of inspection equipment designs for approval. (See 4.4.4). Submit equipment designs as required to Commander, ARRADCOM, ATTN DRDAR-QAA-1, Dover, NJ " 07801

6.4 Drawings. Drawings listed in section 2 of this specification under the heading U.S. Army Armament Research and Development Command (ARRADCOM) may also. include drawings prepared by, and identified as Edgewood Arsenal, Frankfort! Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. " Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Distribution of shop traveler(s), inspection and test results, process controls. Shop traveler(s), inspection and test results (3.7, 4.5.20), and process controls (3.8) shall be sent to: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAA-M, Dover, NJ 07801.

6.6 Contract data requirements.

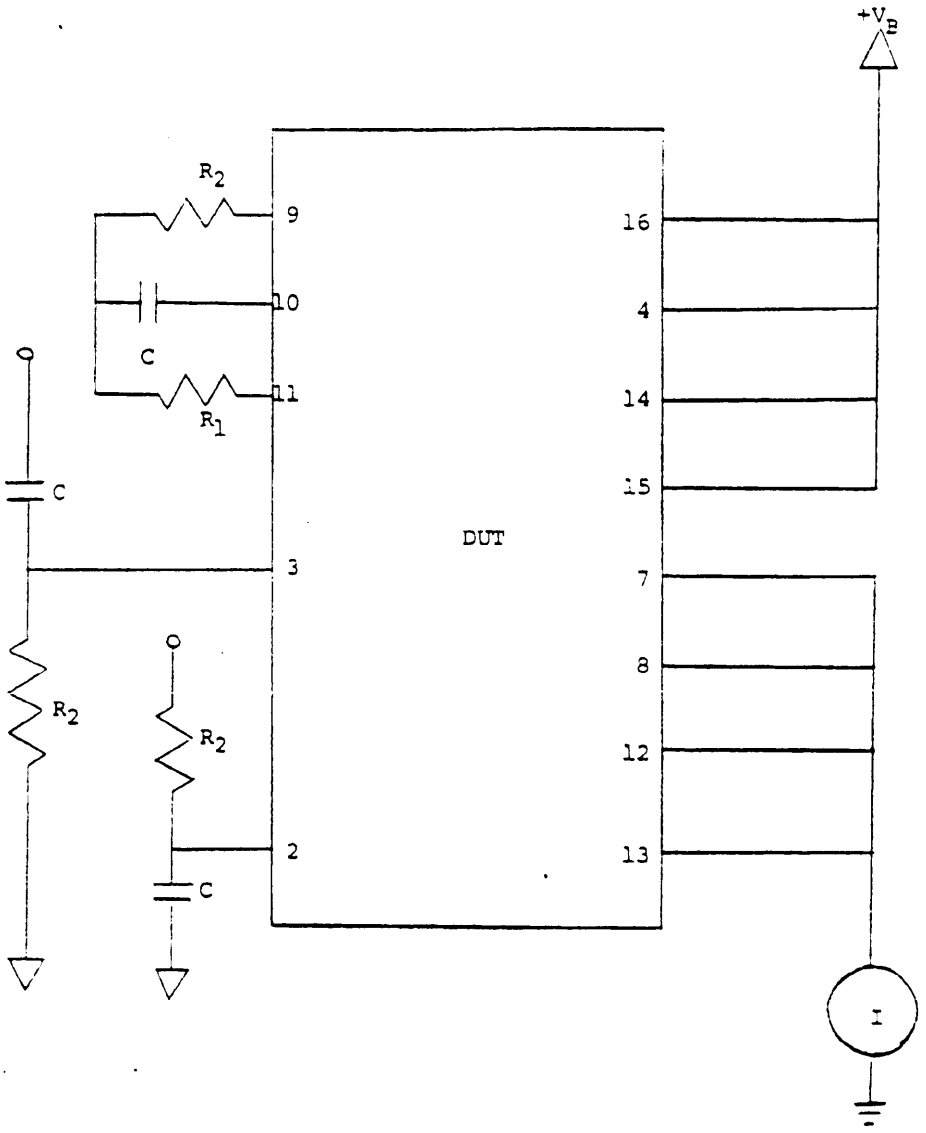
a. Contract data requirements for inspection equipment designs shall conform to data item description DI-E-1118 or DI-I-1714 with the appropriate addendum to each.

b. Shop traveler(s) shall be submitted.

c. Process control documents shall be submitted in accordance with DI-P-1604 (Mod) with attachment i listed on DD Form 1423 incorporated in the contract.

d. Contract data requirements for quality inspection test report in accordance with Data Item Description DI-R-1724 (tailored] giving the detailed test results.

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	NOMINAL VALUE	TOLERANCE	STYLE	TYPE	TCC
C	0.01 uF	± 1.0%	Ceramic	NPO	± 25 ppm
R_1	470 K ohms	± 1.0%	Metal Film		± 50 ppm
R_2	1.0 M ohms	± 5.0%	CC		

FIGURE 1. Oscillator function test configuration

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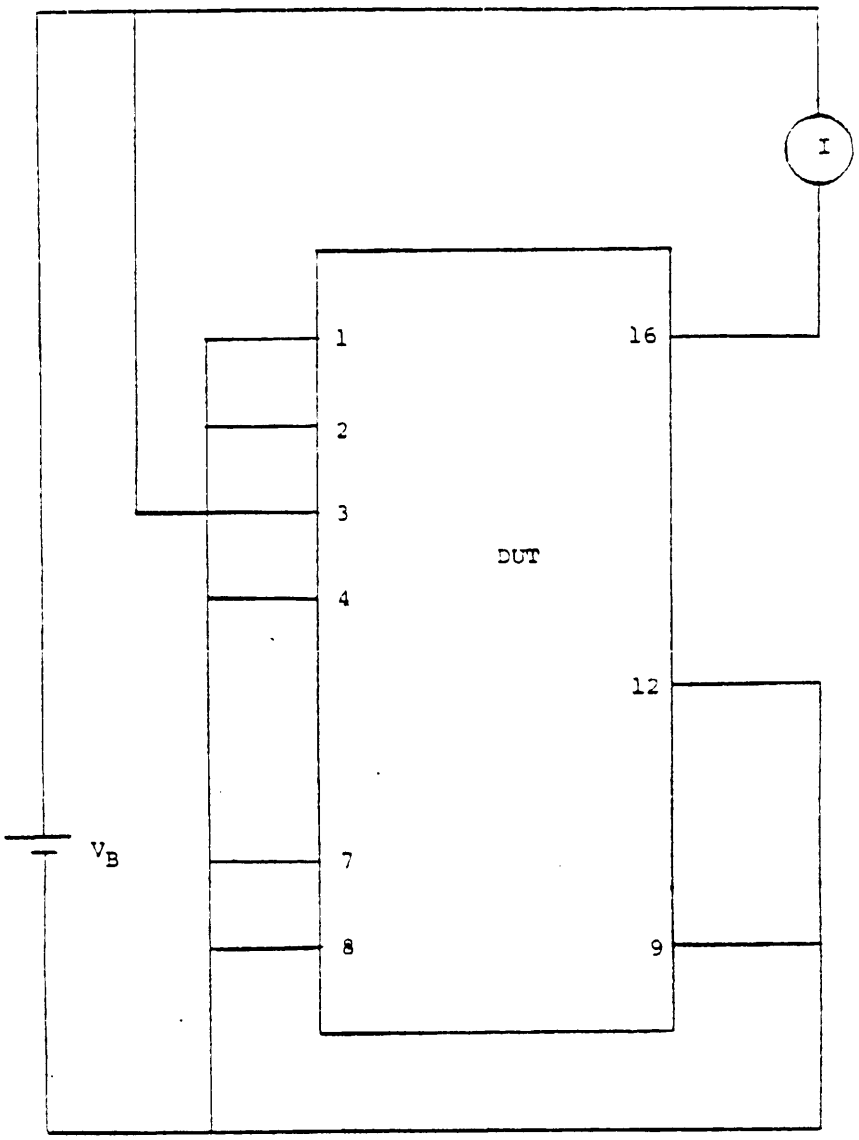


FIGURE 2. Leakage current low test configuration

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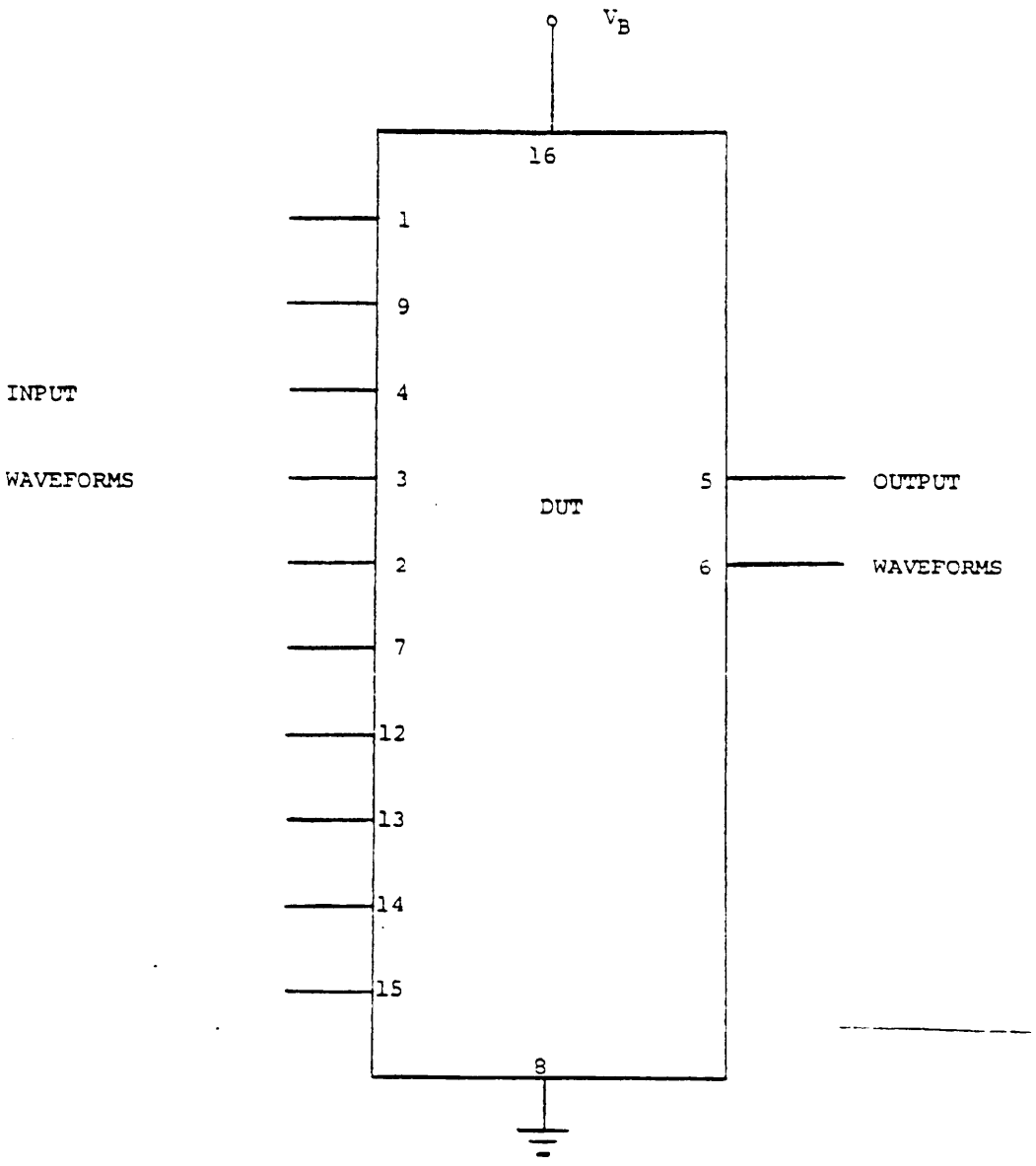


FIGURE 3. Sequential test configuration

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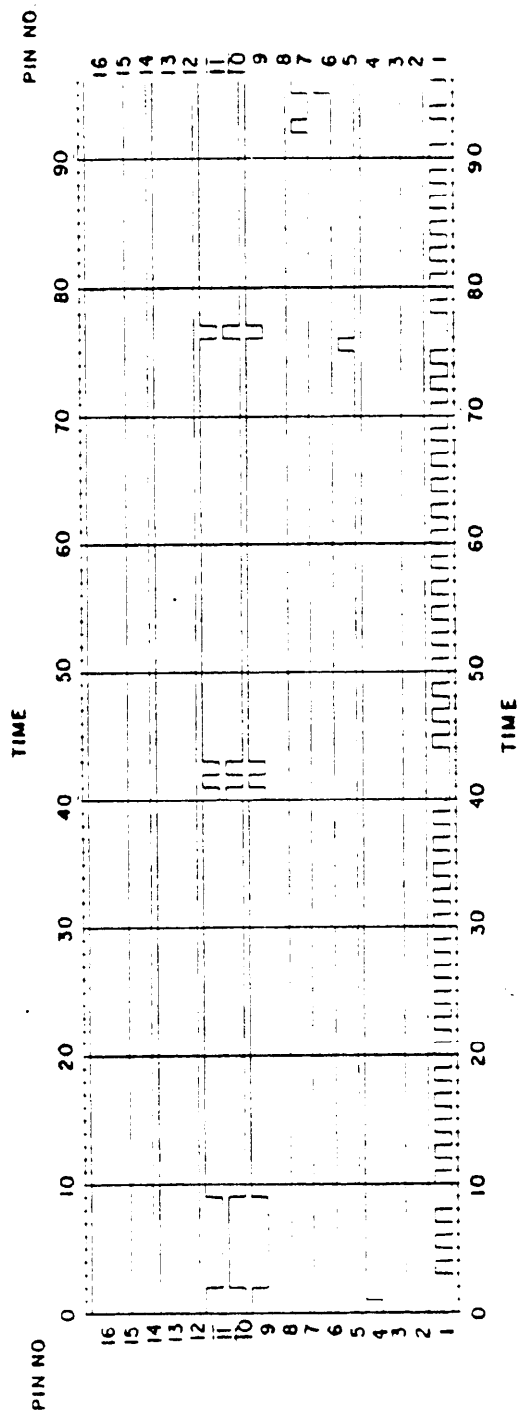


FIGURE 4. Sequential test number 1.

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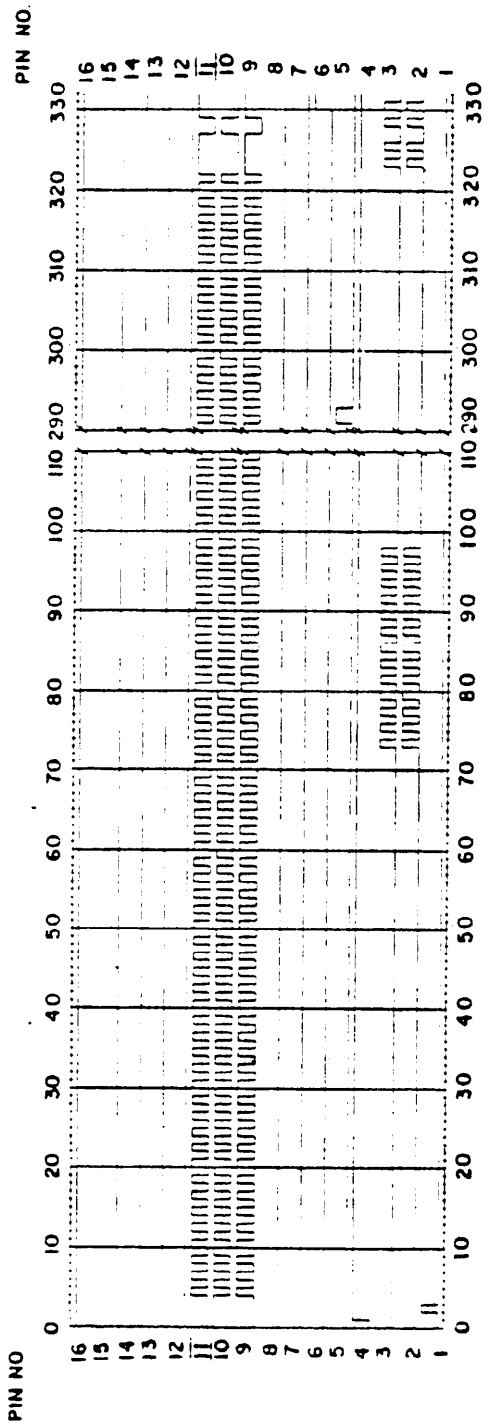


FIGURE 5. Sequential test number 2.

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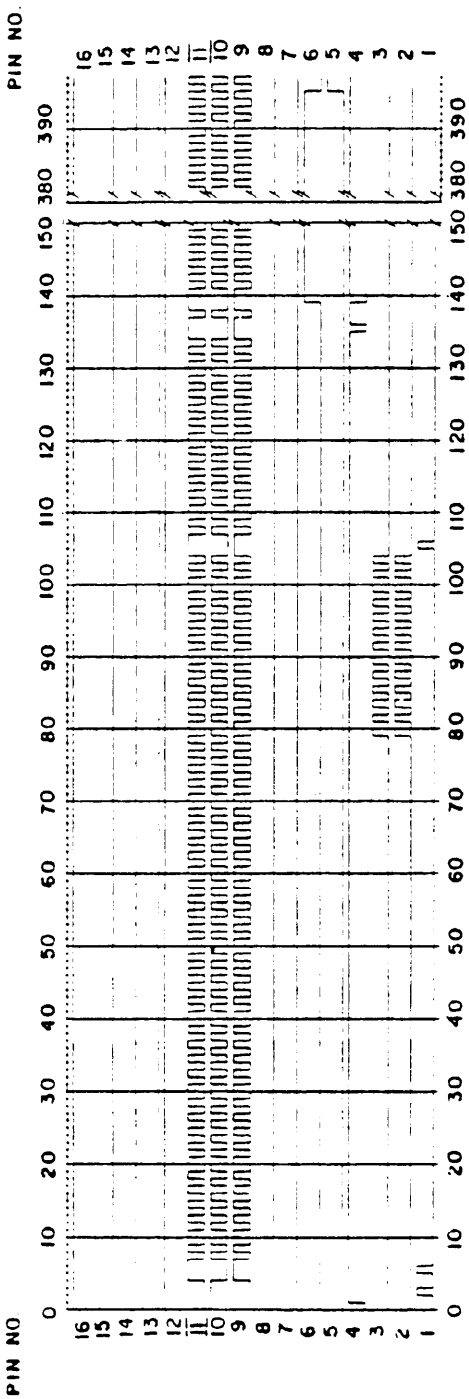


FIGURE 6. Sequential test number 3.

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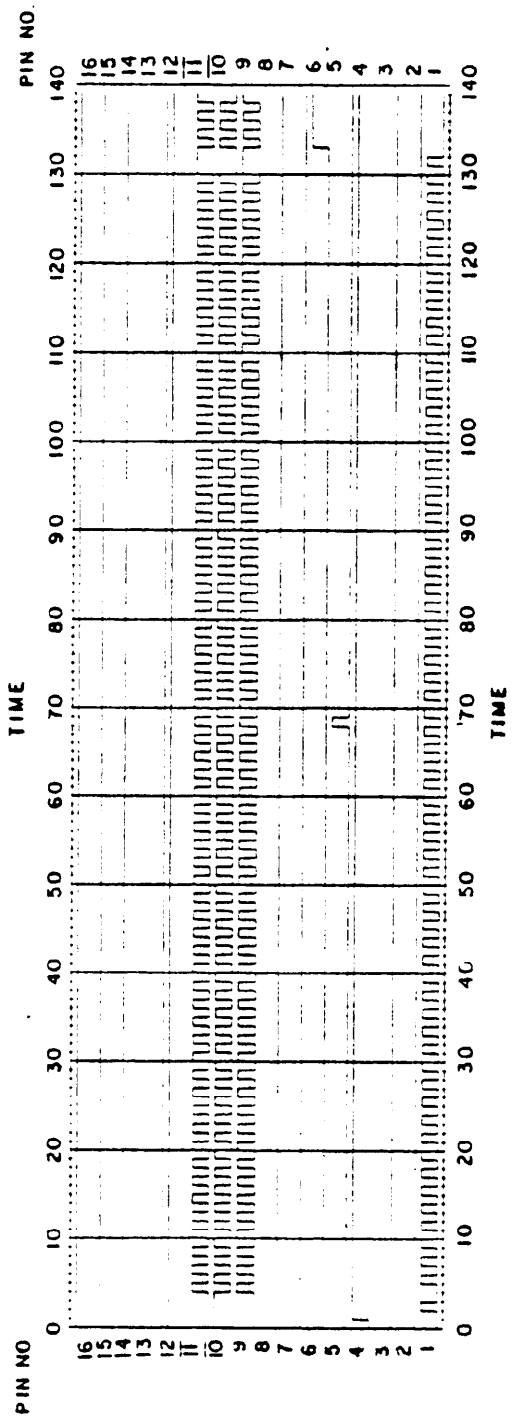


FIGURE 7. Sequential test number 4.

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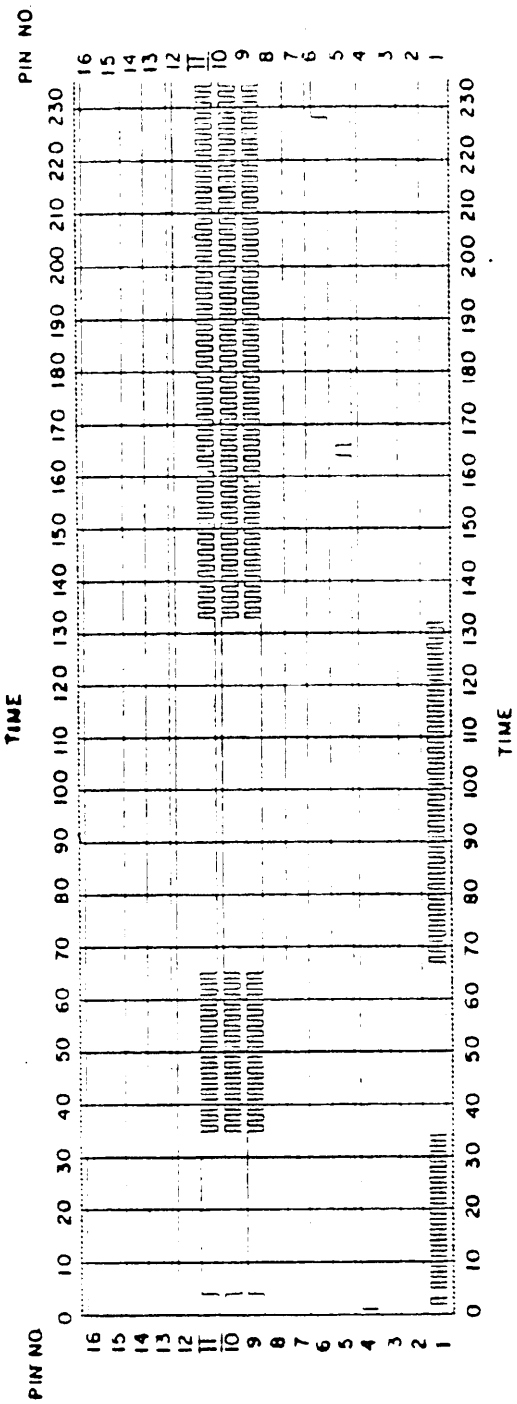


FIGURE 8. Sequential fail test number 5.

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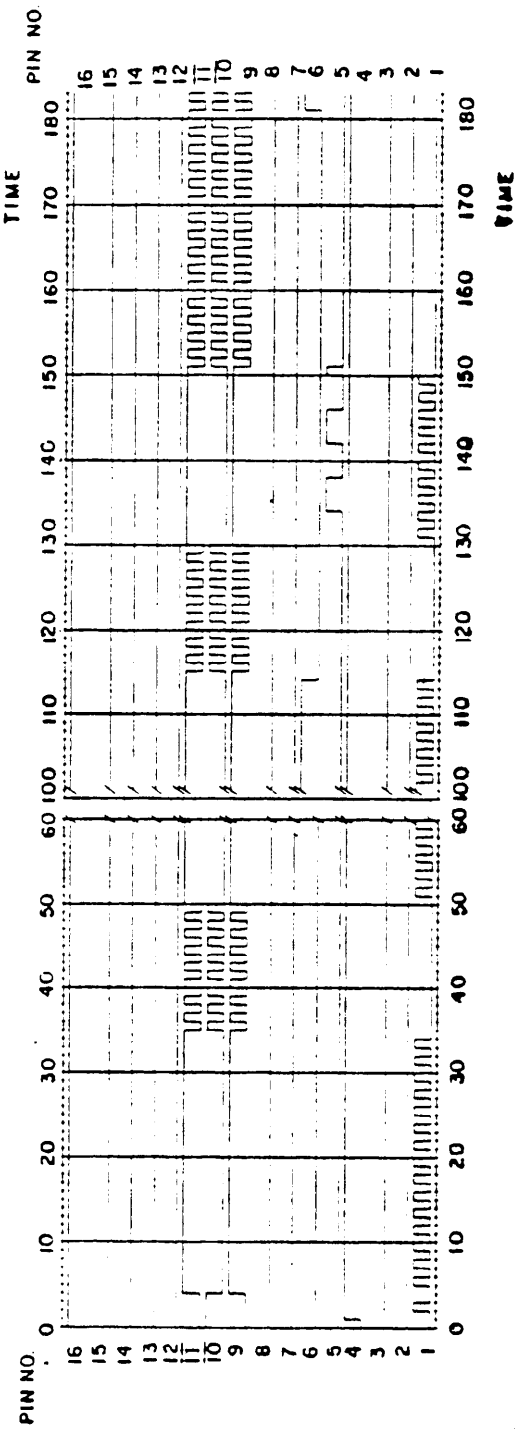


FIGURE 9. Sequential test number 6.

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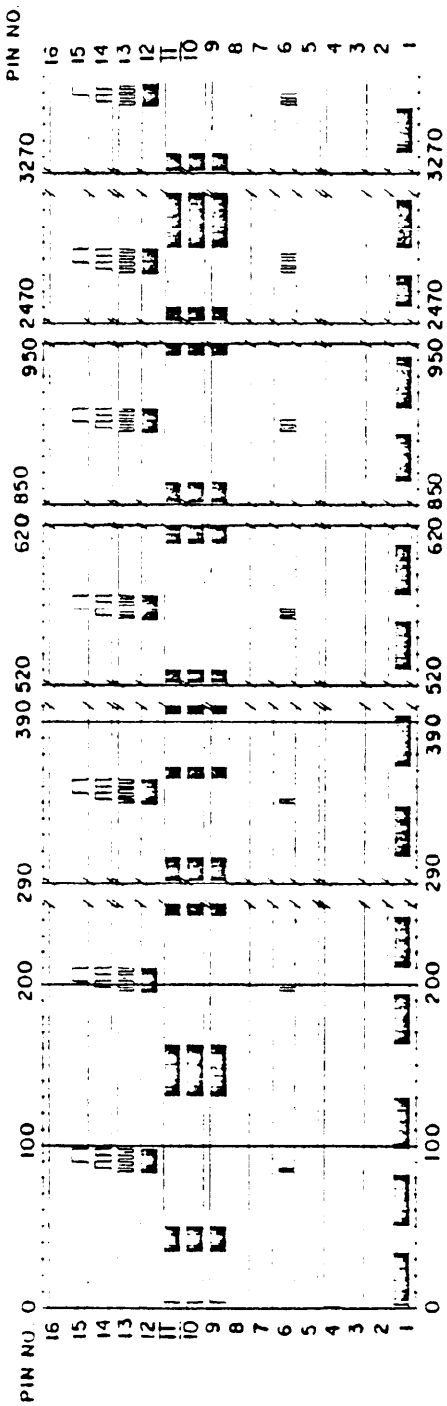
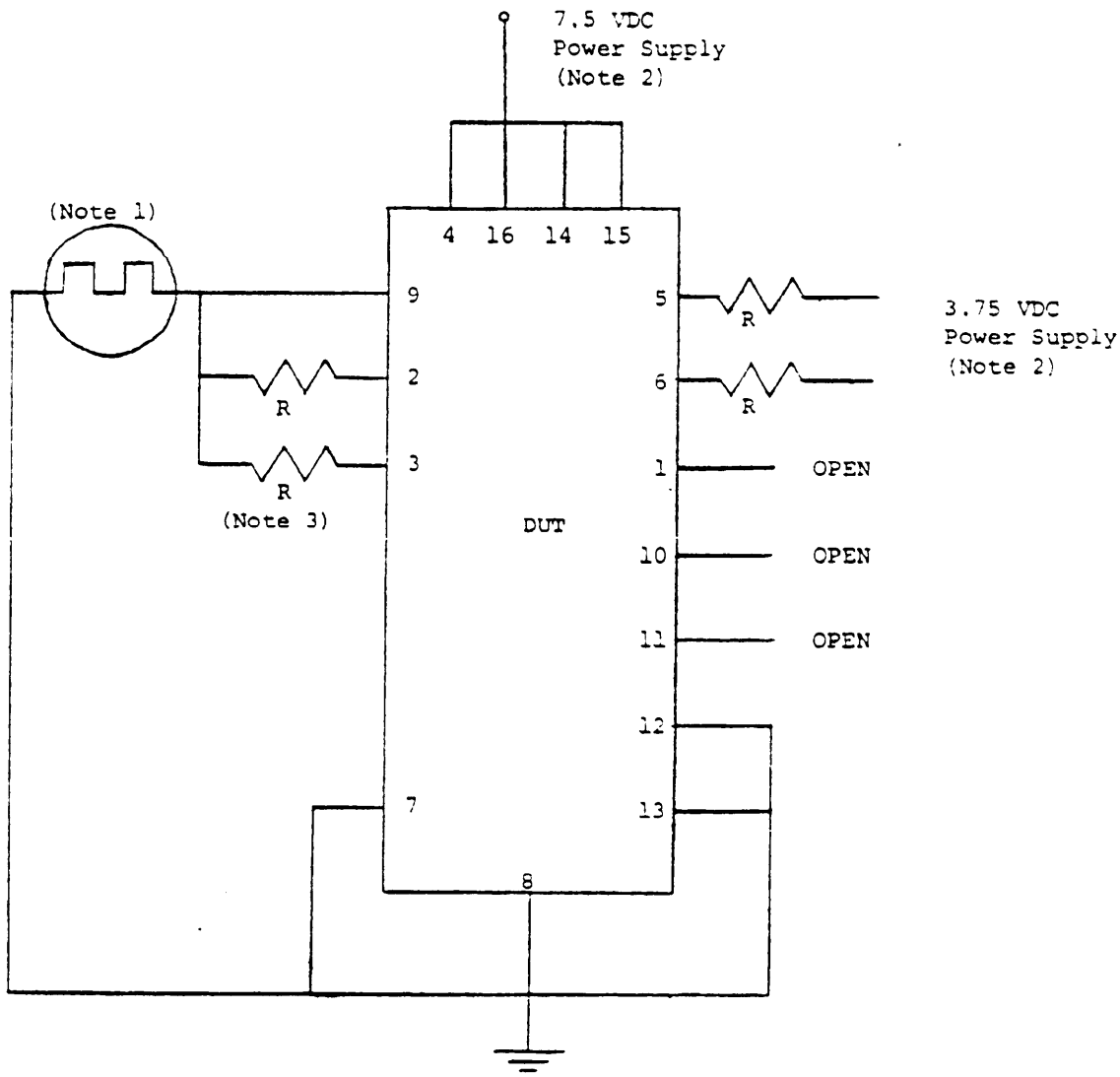


FIGURE 10. Sequential test number 7.

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NOTES:
1. Rack frequency 30 - 70% duty cycle (frequency about 25 KHZ).
2. Both power supplies shall be turned off momentarily at the end of each hour of operation to allow all logic circuits to return to the reset state.
3. All resistors 5K ohms.

FIGURE 11. Dynamic burn-in test configuration

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